Configuring ospf practical

1.Using Cisco Packet Tracer, create a network topology with three routers in two OSPF areas (Area 0 and Area 1), and assign appropriate IP addresses to all interfaces.

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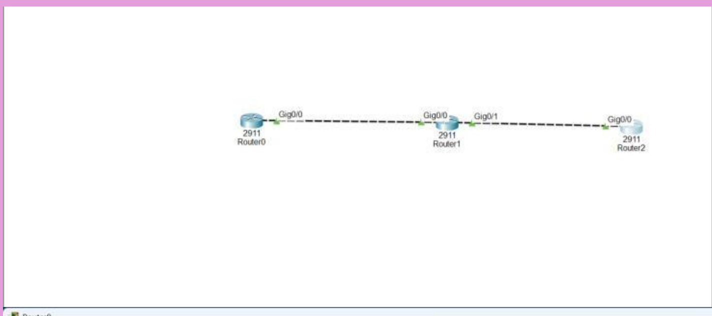
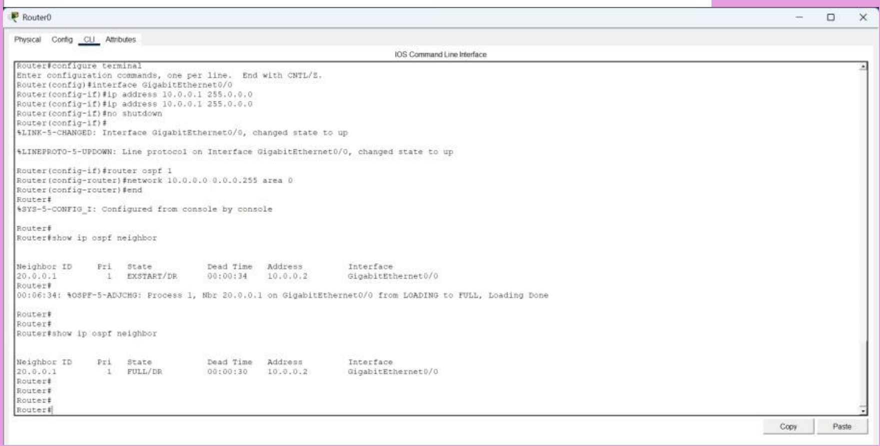
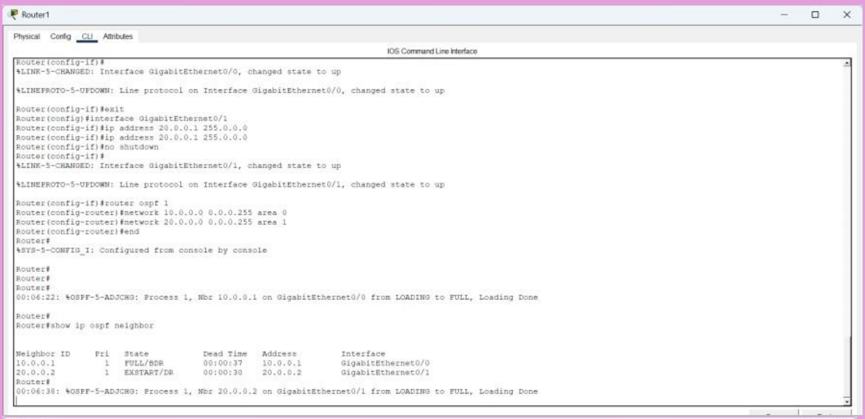
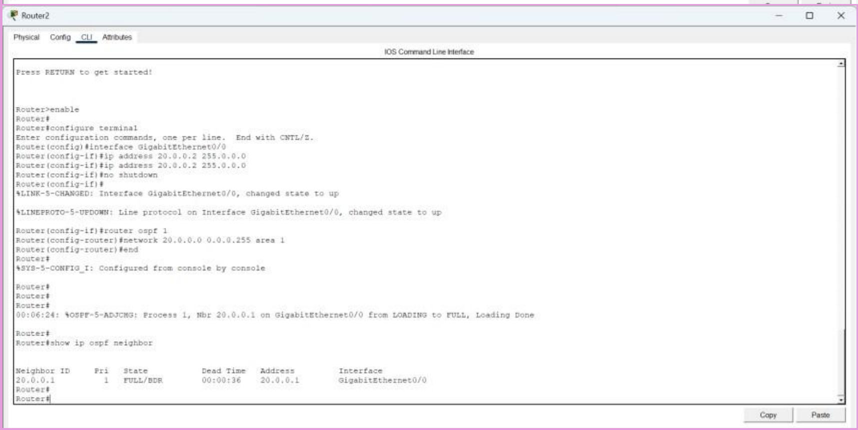
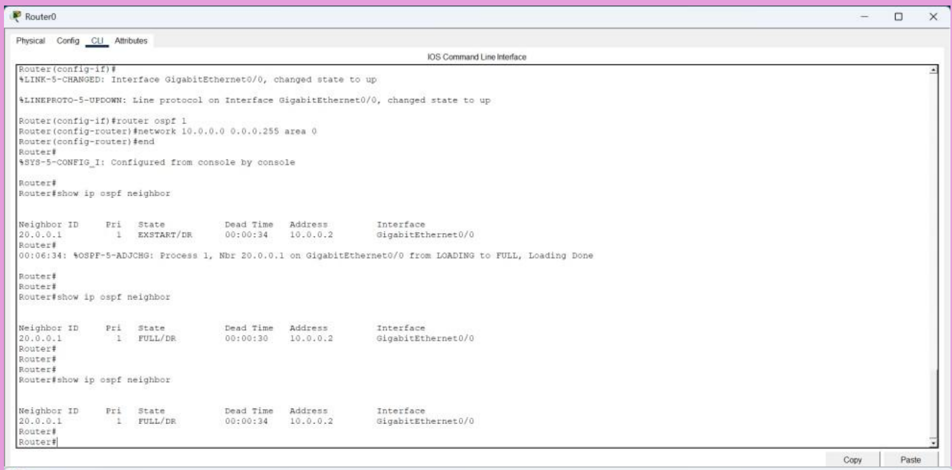








2.Configure OSPF on each router in your topology so that Area 0 acts as the backbone and Area 1 connects through one of the routers; verify OSPF neighbor relationships using the 'show ip ospf neighbor' command. ○


Router0

Physical Config CLI Attributes

IOS Command Line Interface

```
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#router ospf 1
Router(config-router)#network 10.0.0.0 0.0.0.255 area 0
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
Router#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
20.0.0.1          1   EXSTART/DR      00:00:34    10.0.0.2        GigabitEthernet0/0
Router#
00:06:34: %OSPF-5-ADJCHG: Process 1, Nbr 20.0.0.1 on GigabitEthernet0/0 from LOADING to FULL, Loading Done

Router#
Router#
Router#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
20.0.0.1          1   FULL/DR         00:00:30    10.0.0.2        GigabitEthernet0/0
Router#
Router#
Router#
Router#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
20.0.0.1          1   FULL/DR         00:00:34    10.0.0.2        GigabitEthernet0/0
Router#
Router#
```

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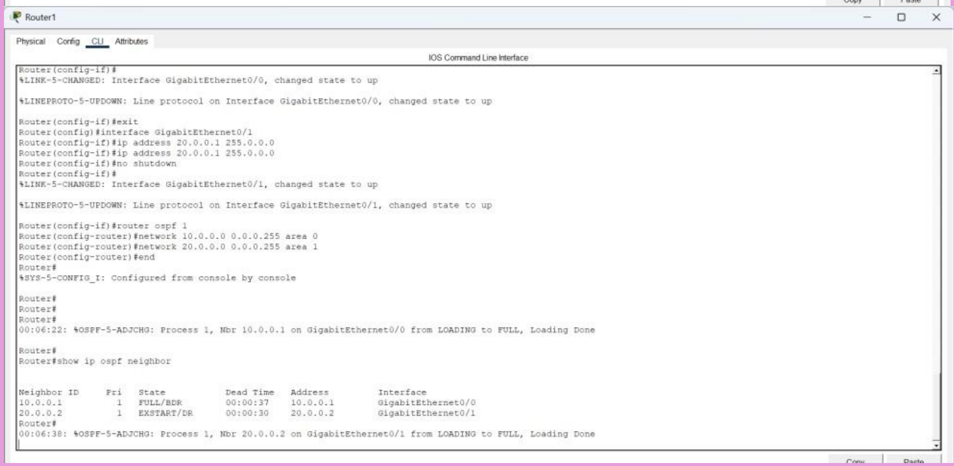
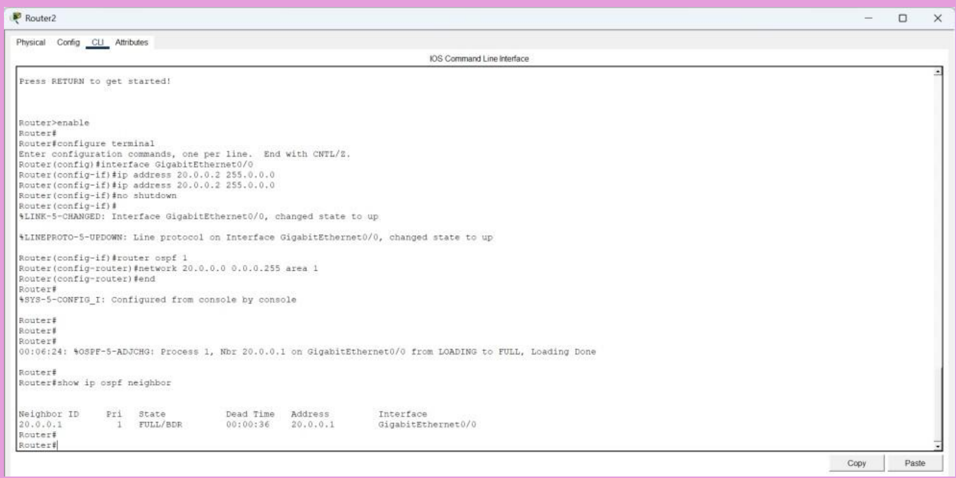

Router1

Physical Config CLI Attributes

IOS Command Line Interface

```
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#interface GigabitEthernet0/1
Router(config-if)#ip address 20.0.0.1 255.0.0.0
Router(config-if)#ip address 20.0.0.1 255.0.0.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config-if)#router ospf 1
Router(config-router)#network 10.0.0.0 0.0.0.255 area 0
Router(config-router)#network 20.0.0.0 0.0.0.255 area 1
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
Router#
Router#
00:06:22: %OSPF-5-ADJCHG: Process 1, Nbr 10.0.0.1 on GigabitEthernet0/0 from LOADING to FULL, Loading Done

Router#
Router#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
10.0.0.1          1   FULL/BDR        00:00:37    10.0.0.1        GigabitEthernet0/0
20.0.0.2          1   EXSTART/DR      00:00:30    20.0.0.2        GigabitEthernet0/1
Router#
00:06:38: %OSPF-5-ADJCHG: Process 1, Nbr 20.0.0.2 on GigabitEthernet0/1 from LOADING to FULL, Loading Done
```


Router2

Physical Config CLI Attributes

IOS Command Line Interface

```
Press RETURN to get started!



Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip address 20.0.0.2 255.0.0.0
Router(config-if)#ip address 20.0.0.2 255.0.0.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#router ospf 1
Router(config-router)#network 20.0.0.0 0.0.0.255 area 1
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
Router#
Router#
00:06:24: %OSPF-5-ADJCHG: Process 1, Nbr 20.0.0.1 on GigabitEthernet0/0 from LOADING to FULL, Loading Done

Router#
Router#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
20.0.0.1          1   FULL/BDR        00:00:36    20.0.0.1        GigabitEthernet0/0
Router#
Router#
```

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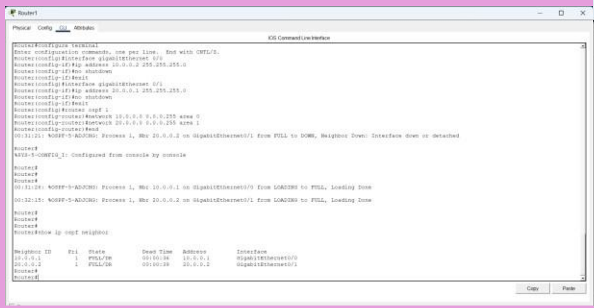
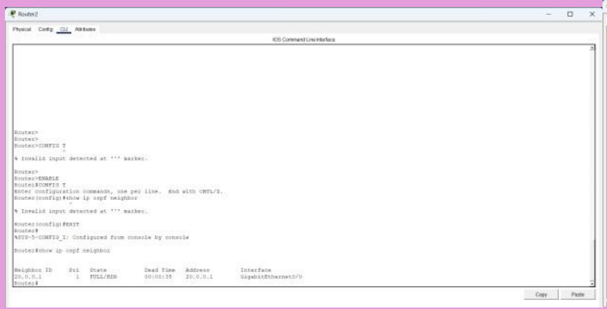
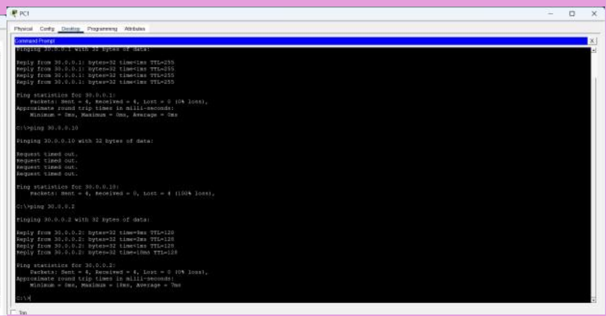
3.Add two PCs to different areas in your OSPF network, assign them IP addresses, and test connectivity by pinging from one PC to the other.







**4.Simulate a router failure by shutting down the interface connecting Area 1 to Area 0, then use 'show ip route' and 'show ip ospf' commands to observe how OSPF reconverges and restores connectivity after re-enabling the interface.

Hint: Track the time it takes for OSPF to update the routing tables after the failure and recovery.**